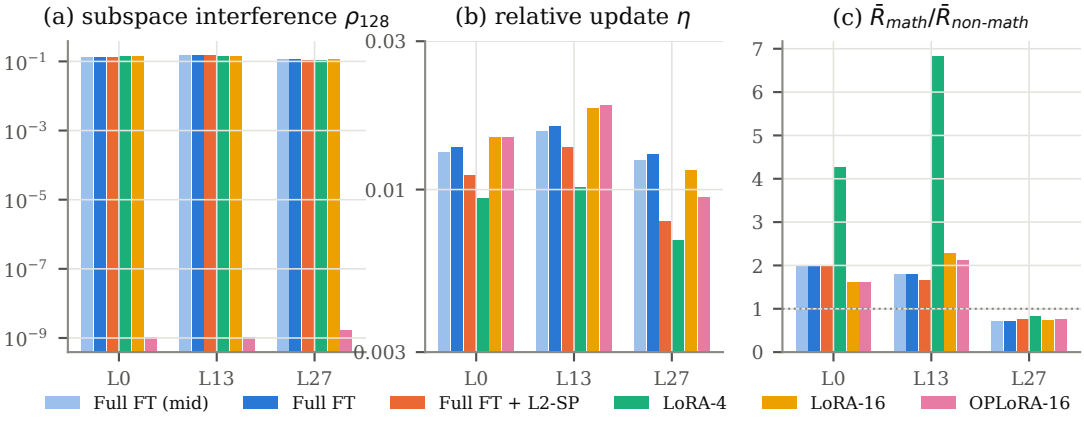
(a) subspace interference ρ_{128}
(b) relative update η
(c) $\bar{R}_{math}/\bar{R}_{non\text{-}math}$
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
0.03
0.01
0.003
7
6
5
4
3
2
1
0
L0
L13
L27
Full FT (mid)
Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16